Pipeline, Labels and Training: WESAD vs. Our Users

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Abstract

The final stress model (ContextGate) is trained in two steps on two different datasets. WESAD (15 lab subjects) teaches the encoder without labels; the survey (STAI-6) answers of our five wearers train the gate and classifier. This document lists the file behind each step, shows how each survey answer becomes a label, states which windows were dropped and why, and describes how the model was trained.

1 Pipelines at a glance

WESAD. Raw files become 7 channels at 1 Hz, which train the encoder without labels. The encoder is then frozen and handed to the user pipeline. WESAD's own labels are used only for the audit.

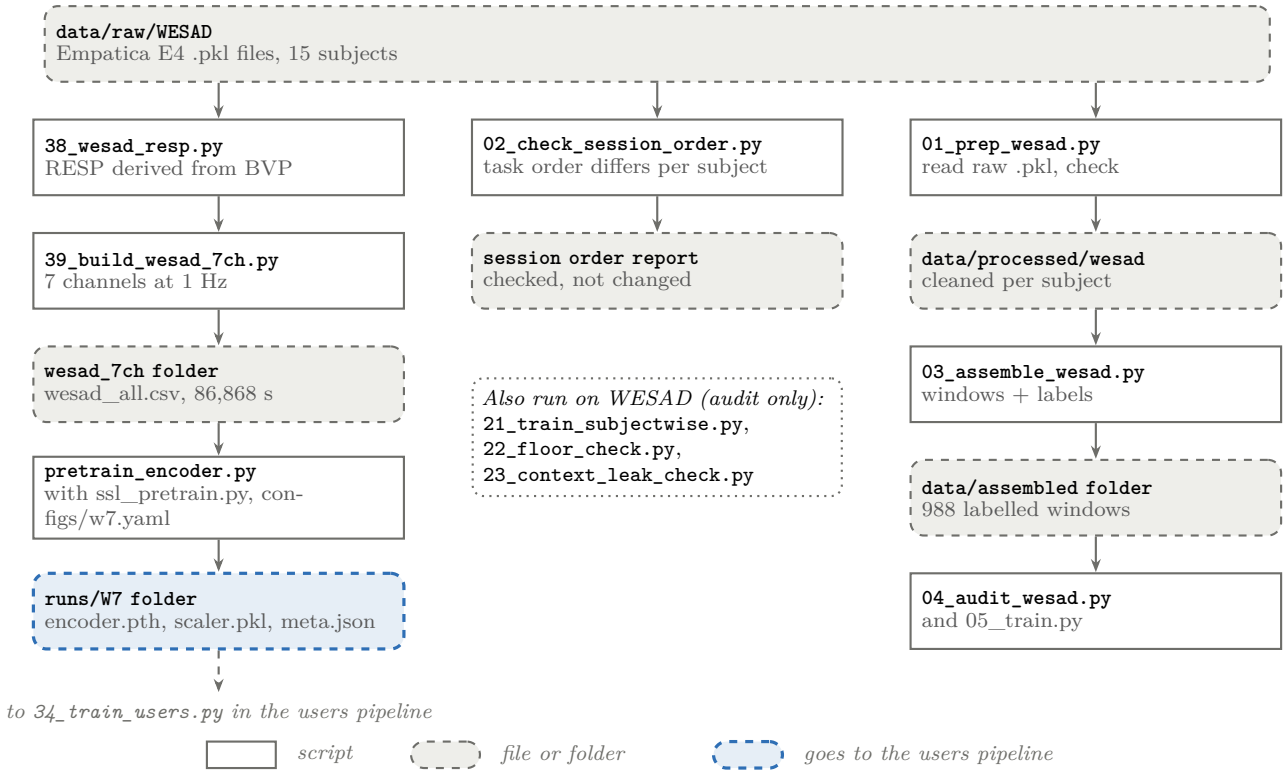


Figure 1: WESAD pipeline, file by file. Left: the path that builds the frozen encoder. Right: the reproduction/audit path that uses WESAD's own labels.

Our users. Three branches (signals, context, labels) are built separately from the raw files and the survey, then meet in the training step. Raw files also feed the context branch through the biomarker

and derivation scripts. The frozen WESAD encoder reads the signals and the stress score is computed for every window.

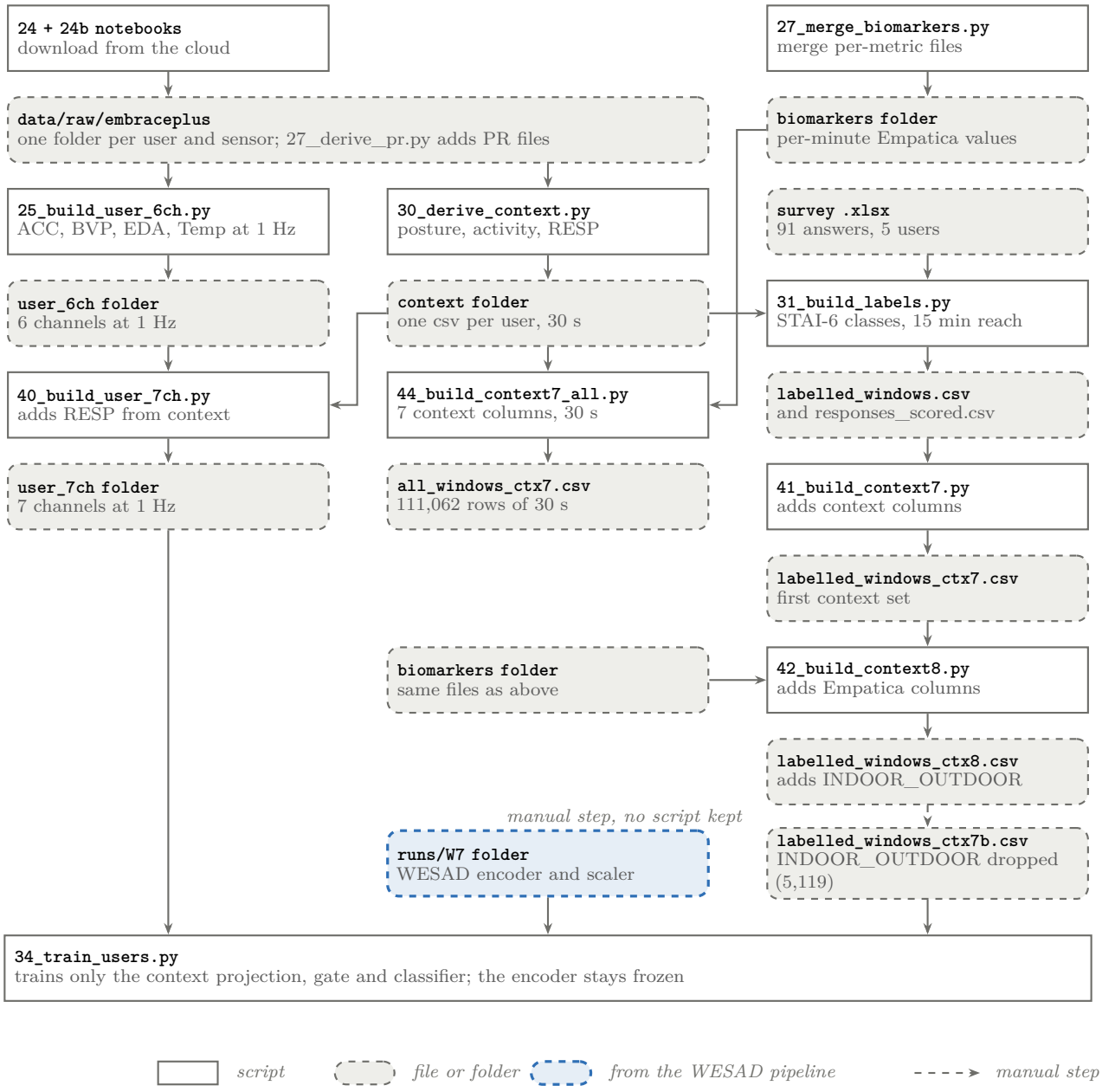


Figure 2: Users pipeline, part 1: from raw recordings and the survey to the training script. Signals (left), context (middle) and labels (right) are built separately and meet in `34_train_users.py`.



Figure 3: Users pipeline, part 2: training output, the stress score for every window, and the checks and demo scripts.

2 The pipeline, stage by stage

WESAD and our wearers share the same 7 channels, the same 60 s windows (stride 30 s) and the same encoder. They differ in who provides labels and context and in what each dataset is used for: WESAD builds the encoder, our users train and test the stress score.

Stage	WESAD (15 lab subjects)	Our users (5 labelled wearers)
1. Raw data	Empatica E4 `.pkl` files. Checked by `s0_check_wesad_raw.py`, `01_prep_wesad.py`, `02_check_session_order.py` (task order differs between subjects)	EmbracePlus raw CSVs, downloaded by `24_download_embraceplus.ipynb` and `24b_download_remaining_users.ipynb`
2. Signals at 1 Hz	`39_build_wesad_7ch.py`: ACC x/y/z, BVP, EDA, Temp from the wrist, 86,868 seconds, 15 subjects	`25_build_user_6ch.py` (ACC x/y/z, BVP, EDA, Temp), pulse rate from `27_derive_pr.py`
3. Breathing channel (RESP)	`38_wesad_resp.py`, derived from BVP, checked against the chest belt ($r = 0.185$)	`40_build_user_7ch.py` adds RESP, output `user_7ch/<user>.csv` (7 channels)
4. Context (7 columns)	Not used in the final model. Only tested for leakage: `23_context_leak_check.py` (Task alone scores 1.000)	`27_merge_biomarkers.py`, `30_derive_context.py`, `44_build_context7_all.py`: HOUR, IS_WEEKEND, POSTURE, ACT_COUNT, ACT_CLASS, BODY_POSITION, SLEEP. Output `all_windows_ctx7.csv` (111,062 rows of 30 s)
5. Labels	Not used in the final model. WESAD's own conditions (baseline = Low, amusement = Medium, stress = High) in `39_build_wesad_7ch.py`, used only for the reproduction and audit	`31_build_labels.py`: STAI-6 answers become Low / Medium / High, attached to windows within 15 min (`labelled_windows.csv`). Then `41_build_context7.py` and `42_build_context8.py` add the context columns, and INDOOR_OUTDOOR is dropped by hand (no script kept) to give `labelled_windows_ctx7b.csv` (5,119 windows)
6. Encoder	`pretrain_encoder.py` and `ssl_pretrain.py` with `configs/w7.yaml`: contrastive, no labels, 2,894 windows. Output `runs/W7/encoder.pth`, `scaler.pkl`, `meta.json`. Then frozen	Not trained. The frozen WESAD encoder and WESAD scaler are applied unchanged
7. Gate and classifier	Only for the audit (`05_train.py`, `21_train_subjectwise.py`, `22_floor_check.py`)	`34_train_users.py`: trains only the context projection, gate and classifier on the labelled windows. Output `runs/FINAL_ctx7_random/model.pth`
8. Stress score	Not produced	`45_score_all_with_context.py`: every window gets $P(\text{Low}), P(\text{Med}), P(\text{High})$ and score $= P(\text{Med}) + 2P(\text{High})$, 0 to 2. Output `runs/SCORES_ctx7/predictions.csv` (104,059 windows)
9. Checks and demo	`04_audit_wesad.py`	`46_context_ablation.py`, `70_actual_vs_detected.py`, then `55_session_checkpoints.py` on the five demo sessions and the Jetson

Table 1: Stage-by-stage file map. The two columns meet at stage 6: the encoder built from WESAD is what the user pipeline reads in stage 7.

3 How a survey answer becomes a label

Each of the 91 answers is scored, put into one of three classes, and then copied onto every 30 s window within 15 minutes either side of the moment it was given. Windows that no answer reaches stay unlabelled and are not used for training. All of this is in `31_build_labels.py`.

Step 1: score the answer (STAI-6). Six items are rated 1 to 4: calm, tense, upset, relaxed, content, worried. Calm, relaxed and content are reversed ($5-\text{rating}$). Add the six to get a raw score of 6 to 24, then multiply by 20/6 to get the STAI-6 score on a 20 to 80 scale.

Step 2: cut into three classes across everyone. The two cut points (46.67 and 56.67) are the 1/3 and 2/3 quantiles of all 91 scores, not of each person's own scores.

Class	STAI-6 score	Raw score	Answers	Windows
Low	up to 46.67	14 or less	33	1,949
Medium	46.67 to 56.67	15 to 16	27	1,567
High	56.67 and above	17 or more	31	1,603

Cutting across everyone means a score of 60 is the same class whoever reported it. WESAD's per-person cut caused S11's calm baseline to become Medium and S14's to become High.

Step 3: attach the answer to windows (±15 minutes). The context file has one row every 30 s for as long as the watch was worn. For each user, every row is matched to the nearest answer in time (`pd.merge_asof`, `direction='nearest'`, tolerance 15 min). If that answer is within 15 minutes, the row gets the answer's score and class; if not, the row gets no label.

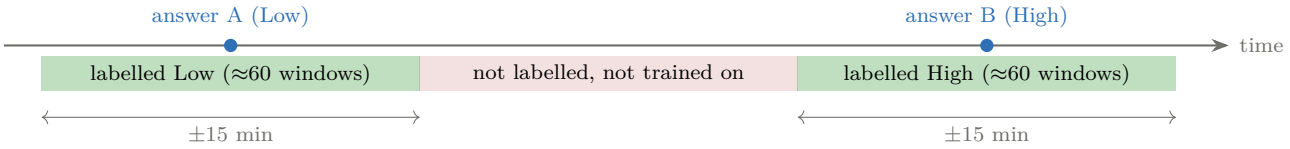


Figure 4: Each answer labels the windows within 15 minutes; windows in the gap stay unlabelled.

- **About 60 windows per answer.** 30 minutes at 30 s steps is 60 rows. The 88 matched answers average 58.2 windows each (range 29 to 60); the shorter ones are probably where the watch was off or the recording started or ended.
- **If two answers are close,** each window takes the nearer one. In practice this almost never happens: the median gap between a person's answers is 244 minutes and only one pair is within 30 minutes.
- **Between answers.** Any window more than 15 minutes from every answer is left unlabelled. It is not guessed, filled in or interpolated, because there is no ground truth. These windows are not used for training, but they are still scored at inference (stage 8): 104,059 windows get a score, only 5,119 carry a label (about 5 %).
- **Why 15 minutes and not more.** The survey asks about the last hour, so one answer describes a stretch of time. A wider reach would make more windows share one label and look like copies of each other rather than separate examples.

Result: 91 answers, 88 matched, 5,119 labelled windows. Per person: user17 13 answers (705 windows), user18 31 (1,649), user25 23 (1,329), user26 18 (1,076), user27 6 (360).

4 How context and signals sit under each label

A labelled window is one row in the labelled file. It carries the label, the 7 context columns and a timestamp. The 60 s of signal underneath it is looked up from the 1 Hz channel file by that timestamp, in `load_windows` inside `34_train_users.py`.

Part	Where it comes from	What the model reads
Label	`31_build_labels.py`: the nearest answer within 15 min	Class 0, 1 or 2 (Low, Medium, High), the training target
Context	`41_build_context7.py` and `42_build_context8.py` add the columns to the labelled windows, then INDOOR_OUTDOOR is dropped (`labelled_windows_ctx7b.csv`). `44_build_context7_all.py` builds the same columns for every 30 s row (`all_windows_ctx7.csv`), used for scoring. Matched on the window's 30 s timestamp	HOUR, IS_WEEKEND, ACT_COUNT as numbers; POSTURE, ACT_CLASS, BODY_POSITION, SLEEP as words. After one-hot encoding this is 19 numbers, fitted on the training rows only
Signal	`user_7ch/<user>.csv`, the 60 rows starting at the window's timestamp	7×60 seconds, scaled with the WESAD scaler, read by the frozen encoder

Context source for each column.

- HOUR and IS_WEEKEND come from the timestamp (IST).
- POSTURE and ACT_COUNT are derived by us from the wrist accelerometer (`30_derive_context.py`).
- ACT_CLASS, BODY_POSITION and SLEEP are Empatica biomarkers, one value per minute, held across the two 30 s steps. Where Empatica recorded nothing the value is the word *Unknown*, so the row is kept.

The label and the context are both attached to the same 30 s timestamp, so the context describes the moment of the window, while the label describes the answer given within 15 minutes of it.

5 What was dropped or ignored, and why

Our users: 5,119 labelled windows start from 111,062 context rows. Most rows are left unlabelled because no answer is near them. Only a few windows and answers are actively thrown away.

What	How many	Why
Windows more than 15 min from every answer	about 105,900 of 111,062 (over 95 %)	No ground truth. Not guessed or interpolated. Not used for training, but scored at inference (104,059 scored windows)
Answers with no window in reach (all user18: 4 Aug 16:16, 5 Aug 14:23, 10 Aug 11:07; all High, 60 to 63)	3 of 91 answers, 88 matched	No context rows within 15 min, so nothing to label. Not forced
Labelled windows without 60 contiguous seconds of signal	about 68 (5,119 down to about 5,051, inferred from split sizes 3,536 + ≈505 + 1,010)	`load_windows` keeps a window only if 60 consecutive seconds exist (span within ±5 s) and it does not run past the end of the file
Missing samples inside a kept window	not dropped	Filled by interpolation; anything still missing after scaling becomes 0
Missing Empatica biomarker (watch off or loose)	not dropped	The cell becomes *Unknown*. The state really is unknown, so the row stays
Context columns removed (not rows)	INDOOR_OUTDOOR, PRV_RMSSD, AROUSAL_AT_REST	INDOOR_OUTDOOR is a survey answer present at under 1 % of windows. PRV_RMSSD had implausible beat timing (about 290 ms against 20 to 50 ms normal). AROUSAL_AT_REST was 0.90 correlated with pulse rate
Survey rows without user id, timestamp, or with an item outside 1 to 4	the script would drop them	The 91 responses used were all valid
The 10 % validation share in the pooled split	about 505 windows	Set aside by the split and not used to choose anything

Our users

What	How many	Why
Meditation and transition seconds (condition labels)	16,136 rows dropped, 30,349 kept	Not baseline, amusement or stress, so a three-class label cannot be made. Only used for the audit
Windows spanning two subjects in pretraining	27 of 2,894	The repository's windowing does not split by subject. Left unchanged and reported. Kept, not dropped
Labelled windows	988, split by subject	The labelled step splits by subject, so a window never mixes two people

WESAD WESAD's labelled windows are not part of the final user model.

6 How the model was trained

Two training steps use two different datasets. The encoder never sees a label. The classifier never sees WESAD.

	Step A: encoder (WESAD)	Step B: gate and classifier (our users)
Script	`pretrain_encoder.py` → `ssl_pretrain.py`, `configs/w7.yaml`	`34_train_users.py`
Data	`wesad_all.csv`: every second of all 15 subjects, 7 channels	`labelled_windows_ctx7b.csv` (5,119 windows) paired with `user_7ch/` signals
Labels	None (self-supervised)	STAI-6 class of the nearest answer within 15 min: Low, Medium, High
Context	None	7 columns, 19 numbers after one-hot, fitted on the training rows only
Signal	2,894 windows of 7×60 s (stride 30 s). WESAD scaler fitted here	7×60 s per window, scaled with that same WESAD scaler
Trained	Conv1d×3 + BiLSTM encoder and projection head; 20 epochs, batch 128, Adam 10^{-3}, NT-Xent ($\tau = 0.5$), loss 4.38 → 3.64	Context projection, gate and classifier only: 177,795 weights; cross-entropy, Adam 10^{-3}, batch 64, 20 epochs
Frozen	Nothing	The encoder (357,760 weights)
Output	`runs/W7/encoder.pth`, `scaler.pkl`, `meta.json`	`runs/FINAL_ctx7_random/model.pth`, `metrics.json`

Contrastive step, in one line. Each window is shown twice with a small random scale (0.9 to 1.1) and noise (0.01). The encoder is trained so the two versions of the same window land close together and different windows land apart.

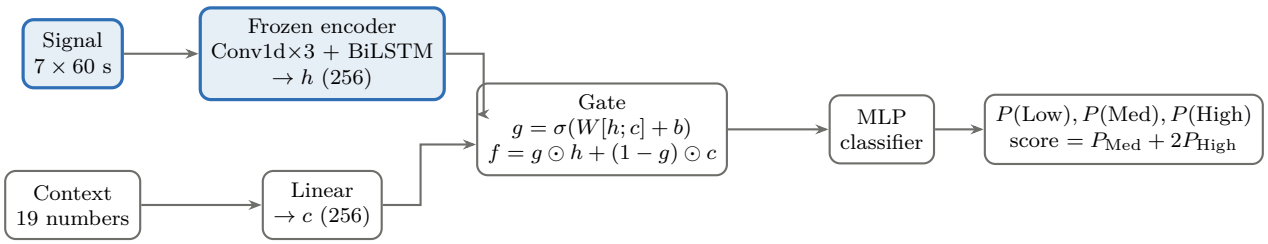


Figure 5: ContextGate for one window. Blue boxes are frozen WESAD components. Trainable: 177,795 weights; total 535,555 (≈2.14 MB).

Model architecture for one window.

How the supervised step is split. The `-split` option chooses the question being asked:

Split	What is held out	Question answered
random (the final saved model)	Random 70/10/20 of windows, 5 seeds	Can it recall? Near-identical windows of one answer fall on both sides, so this is recall
response	Whole survey answers kept on one side	Can it score an answer it has not seen?
user	One person held out, rotating	Does it work on a new person?
temporal	Each person's earlier answers train, later ones test	Can it score the future?

The saved model uses the random split: 3,536 training windows and 1,010 test windows (seed 42), balanced accuracy 0.841 averaged over 5 seeds. That number is *not* evidence of prediction: on answers or people the model has not seen, it does not reliably beat guessing.

WESAD separately. WESAD teaches the encoder what 60 s of wrist signal looks like. Its own labels (baseline, amusement, stress) were used in separate runs only to reproduce and audit the published result: 0.935 on a random split, 0.800 with overlapping windows kept together, and 0.567 for the deep model on unseen subjects. Task as context alone gave 1.000, which is leakage: the task is the session, and the session is the label.

7 Limits of these labels, said plainly

- **One answer is copied onto about 60 windows.** They overlap by 30 s and look alike, so the effective number of independent examples is 88 answers, not 5,119 windows.
- **The answer describes the last hour,** but is attached to a 30-minute stretch centred on the moment it was given.
- **Five people only, and they used the scale differently.** user26 answered Low all 18 times, user18 never answered Low, user27 only Low and High. A model trained on four people does not carry over to the fifth.
- **Classes are cut across everyone.** That keeps a score of 60 meaning the same thing for each person, but a person's own calm baseline can sit in the Medium class.
- **Unlabelled windows are not evidence of calm.** The 105,900 windows between answers have no label at all; they are only scored.